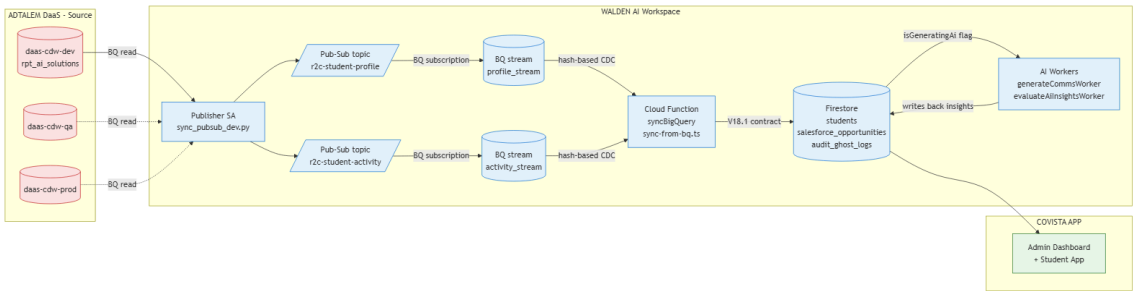


# Covista R2C Pipeline – End-to-End Architecture (May 11)

CDW – Pub/Sub – BigQuery – Firestore – Covista App, plus AI workers.



## Four subscription/event hops

| # | Hop            | Mechanism                                                                                                                                  | Owner                                        |
|---|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1 | CDW – Pub/Sub  | Publisher SA + scheduled BQ read in python-agent/pubsub/sync_pubsub_dev.py                                                                 | Nagendra                                     |
| 2 | Pub/Sub – BQ   | Native BigQuery subscription (no code)                                                                                                     | Nagendra                                     |
| 3 | BQ – Firestore | Cloud Function syncBigQuery() in functions/src/sync-from-bq.ts; trigger via onBqSyncTrigger (Firestore doc write) or scheduled cron        | Nagendra                                     |
| 4 | Firestore – AI | onMessagePublished workers (generateCommsWorker, evaluateAiInsightsWorker) in functions/src/index.ts; AI results written back to same docs | Jaishir (AI gen) + Nagendra (PubSub workers) |

## Identity / IAM summary

- **Today (dev):** runs as my `d51029691-c@mail.waldenu.edu` user via Cloud Shell (reads CDW dev/qa/prod, publishes to Walden topics – verified 5/11).
- **Target (prod):** dedicated SA `pubsub-cdw-publisher@dev-wu-agenticaai-app-proj.iam.gserviceaccount.com` with:
  - `roles/bigquery.dataViewer` on `rpt_ai_solutions` in `daas-cdw-{dev,qa,prod}`
  - `roles/bigquery.jobUser` on publisher project
  - `roles/pubsub.publisher` on Walden topics
- BQ – Firestore leg already runs as the Firebase Admin SA (no new IAM needed).